C. Equal Frequencies

time limit per test: 2 seconds
memory limit per test: 512 megabytes

Let's call a string *balanced* if all characters that are present in it appear the same number of times. For example, "`coder`", "`appall`", and "`ttttttt`" are balanced, while "`wowwow`" and "`codeforces`" are not.

You are given a string s of length n consisting of lowercase English letters. Find a balanced string t of the same length n consisting of lowercase English letters that is different from the string s in as few positions as possible. In other words, the number of indices i such that $s_i \neq t_i$ should be as small as possible.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

Each test case consists of two lines. The first line contains a single integer n ($1 \leq n \leq 10^5$) — the length of the string s.

The second line contains the string s of length n consisting of lowercase English letters.

It is guaranteed that the sum of n over all test cases does not exceed 10^5.

Output

For each test case, print the smallest number of positions where string s and a balanced string t can differ, followed by such a string t.

If there are multiple solutions, print any. It can be shown that at least one balanced string always exists.

Example

input

```
4
5
hello
10
codeforces
5
eevee
6
appall
```

output

```
1
helno
2
codefofced
1
eeeee
0
appall
```

Note

In the first test case, the given string "`hello`" is not balanced: letters '`h`', '`e`', and '`o`' appear in it once, while letter '`l`' appears twice. On the other hand, string "`helno`" is balanced: five distinct letters are present in it, and each of them appears exactly once. Strings "`hello`" and "`helno`" differ in just one position: the fourth character. Other solutions are possible too.

In the second test case, string "`codefofced`" is balanced since only letters '`c`', '`o`', '`d`', '`e`', and '`f`' are present in it, and each of them appears exactly twice.

In the third test case, string "`eeeee`" is balanced since only letter '`e`' is present in it.

In the fourth test case, the given string "`appall`" is already balanced.